**Ex.No.:06**

## Capturing and Analyzing the Packet Traces of TCP

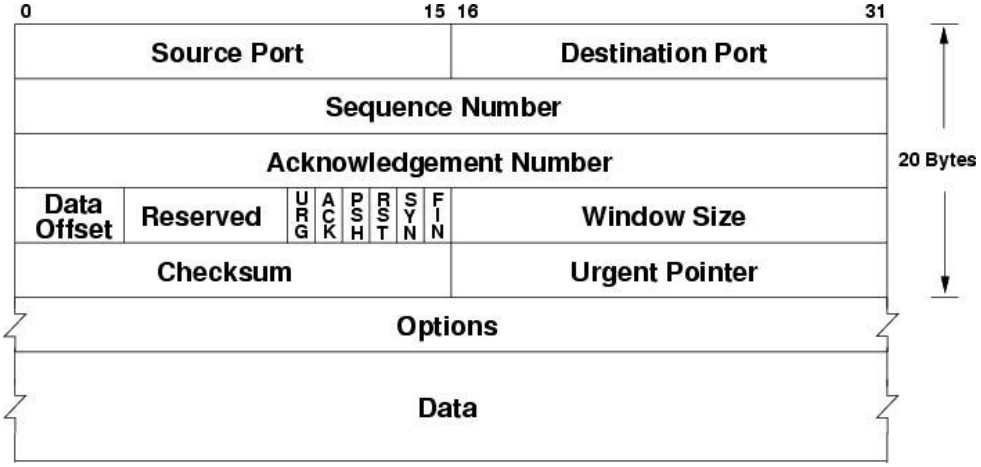
**Aim:**

To capture and analyze the traces of TCP packets using Wireshark tool

**TCP:**

- TCP meaning **Transmission Control Protocol**, is a communications standard for delivering data and messages through networks
- It lies between the Application and Network Layers which are used in providing **reliable delivery services**
- It is a **connection-oriented protocol** for communications that helps in the exchange of messages between different devices over a network

**TCP packet structure**




Some of the most prominent features of Transmission control protocol are

**1. Segment Numbering System**

- TCP keeps track of the segments being transmitted or received by assigning numbers to each and every single one of them
- A specific Byte Number is assigned to data bytes that are to be transferred while segments are assigned sequence numbers
- Acknowledgment Numbers are assigned to received segments
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**2. Connection Oriented**

- It means sender and receiver are connected to each other till the completion of the process
- The order of the data is maintained i.e. order remains same before and after transmission

**3. Full Duplex**

- In TCP data can be transmitted from receiver to the sender or vice – versa at the same time
- It increases efficiency of data flow between sender and receiver

**4. Flow Control**

- Flow control limits the rate at which a sender transfers data
- This is done to ensure reliable delivery
- The receiver continually hints to the sender on how much data can be received (using a sliding window)

**5. Error Control**

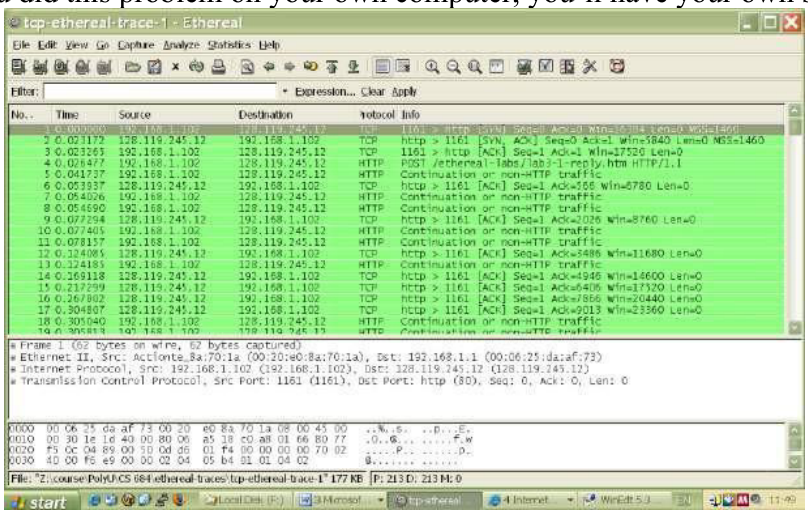
- TCP implements an error control mechanism for reliable data transfer
- Error control is byte-oriented
- Segments are checked for error detection
- Error Control includes – Corrupted Segment & Lost Segment Management, Out-of-order segments, Duplicate segments, etc

**6. Congestion Control**

- TCP takes into account the level of congestion in the network
- Congestion level is determined by the amount of data sent by a sender

1. What is the IP address and TCP port number used by your client computer (source) to transfer the file to gaia.cs.umass.edu?
2. What is the IP address and port number used by gaia.cs.umass.edu to receive the file.
   **Solution:** Client computer (source) IP address: 192.168.1.102 TCP port number: 1161

   Destination computer: gaia.cs.umass.edu IP address: 128.119.245.12
   TCP port number: 80
3. If you did this problem on your own computer, you'll have your own solution

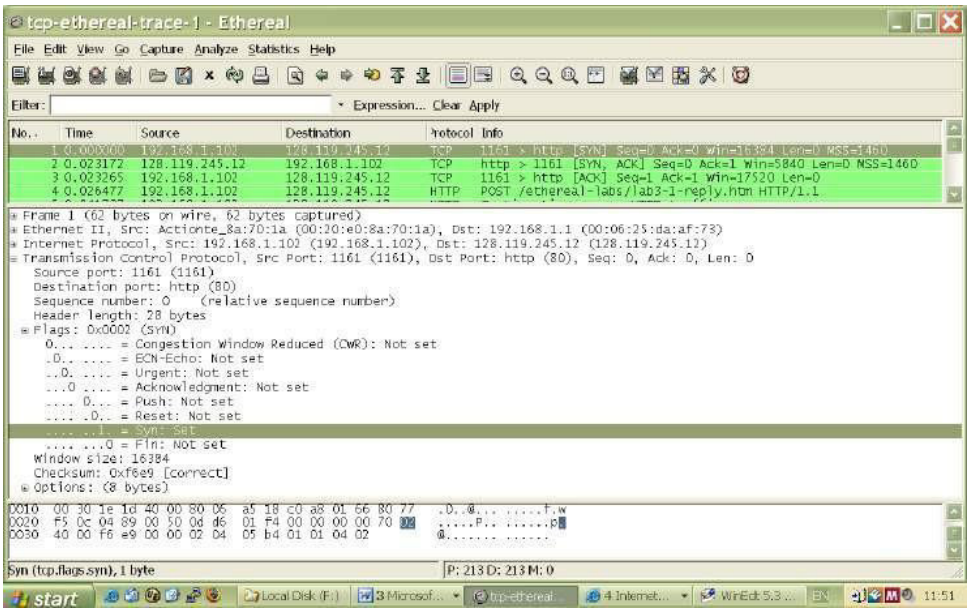



**IP addresses and TCP port numbers of the client computer (source) and gaia.cs.umass.edu**

4. What is the sequence number of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu? What is it in the segment that identifies the segment as a SYN segment?

**Solution:** Sequence number of the TCP SYN segment is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu. The value is 0 in this trace.

The SYN flag is set to **1** and it indicates that this segment is a SYN segment.

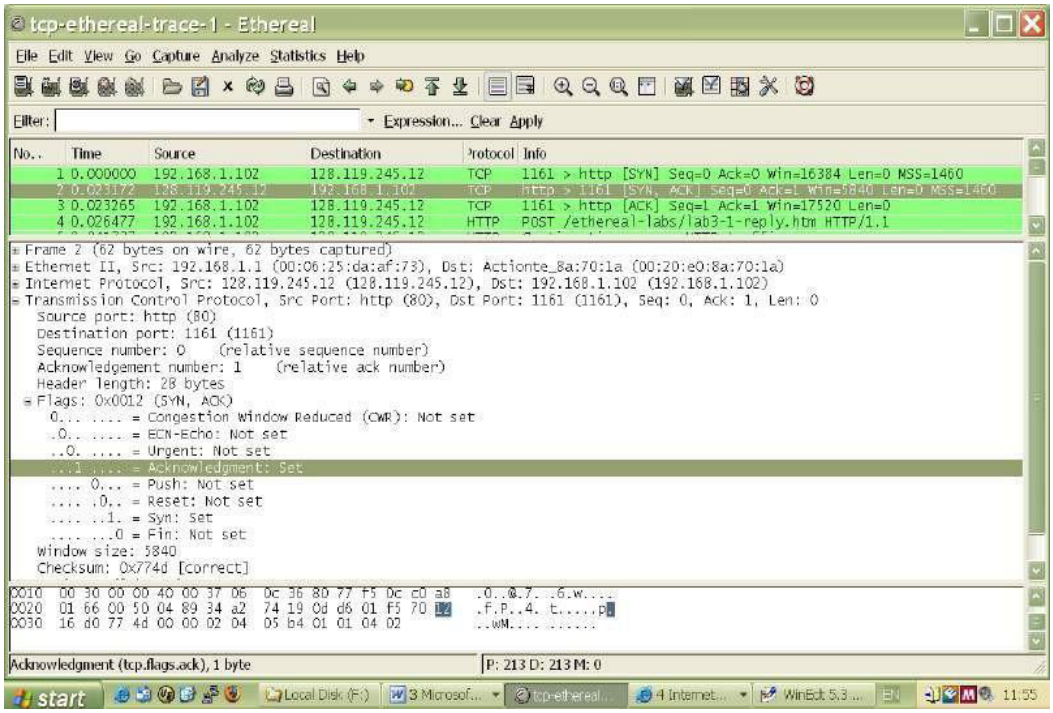



5. What is the sequence number of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is the value of the ACKnowledgement field in the SYNACK segment? How did gaia.cs.umass.edu determine that value? What is it in the segment that identifies the segment as a SYNACKsegment?

**Solution:** Sequence number of the SYNACK segment from gaia.cs.umass.edu to the client computer in reply to the **SYN** has the value of **0** in this trace.

The value of the **ACK**nowledgement field in the **SYNACK** segment is **1**. The value of the ACKnowledgement field in the SYNACK segment is determined by gaia.cs.umass.edu by adding 1 to the initial sequence number of SYN segment from the client computer (i.e. the sequence number of the SYN segment initiated by the client computer is 0.).

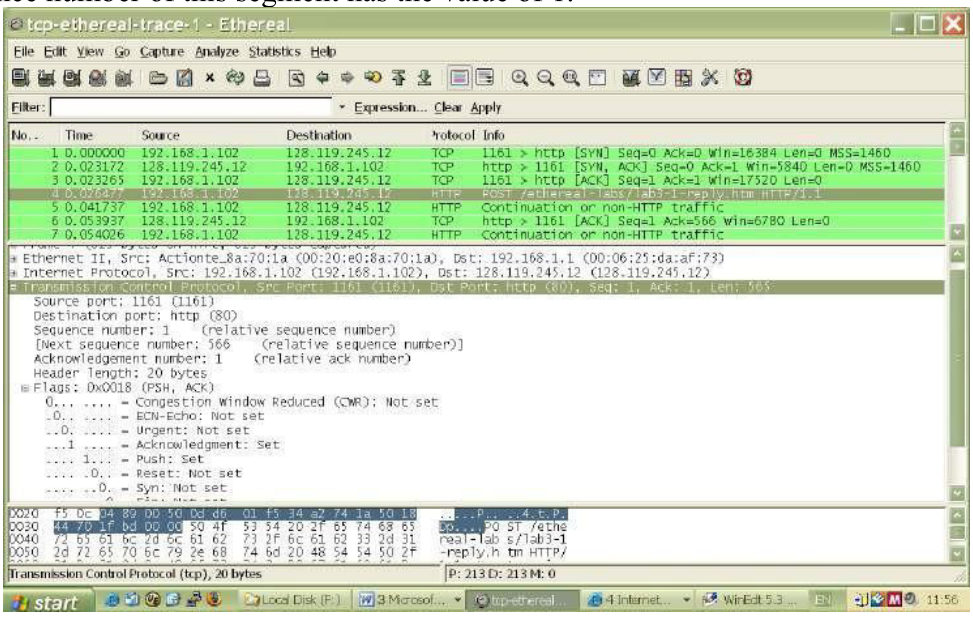
The SYN flag and Acknowledgement flag in the segment are set to **1** and they indicate that this segment is a SYNACK segment.




**Sequence number and Acknowledgement number of the SYNACK segment**

6. What is the sequence number of the TCP segment containing the HTTP POST command?

**Solution:** No. 4 segment is the TCP segment containing the HTTP POST command. The sequence number of this segment has the value of 1.




**Sequence number of the TCP segment containing the HTTP POST command**

7. What is the minimum amount of available buffer space advertised at the received for the entire trace? Does the lack of receiver buffer space ever throttle the sender?

**Solution:** The minimum amount of buffer space (receiver window) advertised at gaia.cs.umass.edu for the entire trace is **5840 bytes**, which shows in the first acknowledgement from the server. This receiver window grows steadily until a maximum receiver buffer size of **62780 bytes**. The sender is never throttled due to lacking of receiver buffer space by inspecting this trace.

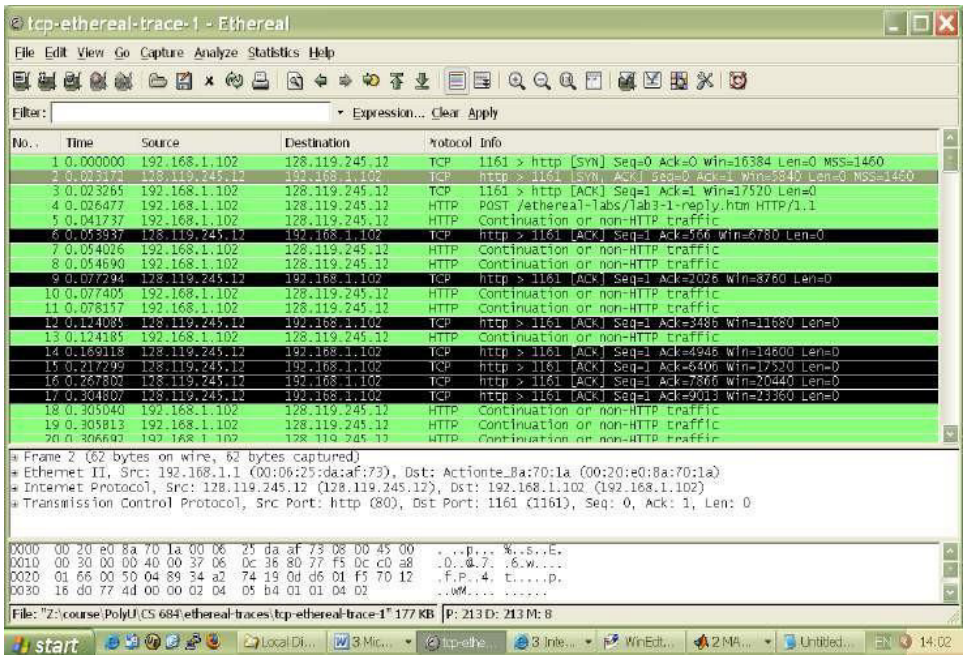



**Figure 9: Minimum receive window advertised at gaia.cs.umass.edu (packet No. 2)**

**Result:**

Thus, capturing and analyzing TCP packets using Wireshark has been completed successfully.